

RESEARCH ON TRANSLATIONS OF TESTS

Depression Anxiety Stress Scales (DASS-21) for Finnish-Speaking Adults: Validation and Normative Data

Luka E. J. Vähäsarja¹, Jari O. Lipsanen^{2, 3}, Anne M. Kouvonen^{4, 5}, Eero T. Lahelma¹,
Raimo Lappalainen⁶, Marianna Virtanen^{7, 8}, and Tea M. Lallukka¹

¹ Department of Public Health, University of Helsinki

² Department of Psychology and Logopedics, University of Helsinki

³ The Finnish Psychological Association, Helsinki, Uusimaa, Finland

⁴ Faculty of Social Sciences, University of Helsinki

⁵ Centre for Public Health, Queen's University Belfast

⁶ Department of Psychology, University of Jyväskylä

⁷ School of Educational Sciences and Psychology, University of Eastern Finland

⁸ Division of Insurance Medicine, Department of Clinical Neuroscience, Karolinska Institute

We conducted the first validation of the Depression Anxiety Stress Scales (DASS-21) in Finnish. DASS-21 is a short public domain questionnaire, which presents a way to quickly and effectively screen for mental ill health. We recruited two large samples, one aged 24–45 ($N = 3,101$ [2,488 women]), and the other aged 60–82 ($N = 5,462$ [4,473 women]), all employees of the city of Helsinki at inclusion (2017 and 2000–2002). DASS-21 measured depression, anxiety, stress, and general distress reliably among Finnish-speaking adults. It appeared invariant with age and gender as evinced by invariance analyses, latent mean comparisons, and an examination of psychometric properties for the subscales and individual items. The subscales negatively correlated with the Emotional Well-being subscale of the RAND-36, as expected. A comparison of five structural models using confirmatory factor analyses and a robust estimation method (weighted-least-squares method) showed a good fit for a one-factor solution. We discuss the use and interpretation of the DASS-21 as both a measure of specific affective symptoms and unidimensional general distress. We provide future researchers and clinicians with norms and estimates of measurement error among Finnish-speaking adults.

Public Significance Statement

We evaluated the Finnish translation of the Depression Anxiety Stress Scales with one of the largest samples collected on the Depression Anxiety Stress Scales in any language so far. Clinicians and researchers may use our normative data for Finnish speakers and no longer rely on norms from English-speaking countries.

Keywords: depression, anxiety, stress, validity, invariance

Supplemental materials: <https://doi.org/10.1037/pas0001355.supp>

Javier Suárez-Álvarez served as action editor.

Luka E. J. Vähäsarja  <https://orcid.org/0009-0007-9474-3831>

Jari O. Lipsanen  <https://orcid.org/0000-0003-0746-2745>

Anne M. Kouvonen  <https://orcid.org/0000-0001-6997-8312>

Eero T. Lahelma  <https://orcid.org/0000-0002-1064-1333>

Raimo Lappalainen  <https://orcid.org/0000-0001-6683-134X>

Marianna Virtanen  <https://orcid.org/0000-0001-8361-3301>

Tea M. Lallukka  <https://orcid.org/0000-0003-3841-3129>

This study was supported by the Research Council of Finland (Grant 330527) and the Kela—Social Insurance Institution of Finland (Grant 29/26/2020) awarded to Tea M. Lallukka.

Luka E. J. Vähäsarja played a lead role in conceptualization,

formal analysis, software, writing—original draft, and writing—review and editing. Jari O. Lipsanen played an equal role in writing—review and editing. Anne M. Kouvonen played an equal role in writing—review and editing. Eero T. Lahelma played an equal role in writing—review and editing. Raimo Lappalainen played an equal role in writing—review and editing and translation. Marianna Virtanen played an equal role in funding acquisition, project administration, and supervision, a supporting role in conceptualization, and an equal role in writing—review and editing.

Correspondence concerning this article should be addressed to Luka E. J. Vähäsarja, Department of Public Health, University of Helsinki, P.O. Box 20 FIN 00014, Finland. Email: luka.vahasarja@aalto.fi

Depression and anxiety disorders constitute the greatest burden of mental disorders, both globally (Global Burden of Disease 2019 Mental Disorders Collaborators, 2022) and in Finland (Finnish Institute for Health and Welfare, 2023). They are also the greatest cause behind disease-adjusted years lost (Global Burden of Disease 2019 Mental Disorders Collaborators, 2022). Despite the differences between the two disorders, they frequently co-occur and exhibit longitudinal interactions (Shankman & Klein, 2003; Winer et al., 2017). Understanding depression and anxiety requires robust, standardized instruments with a broad scope of mental health.

Depression Anxiety Stress Scales (DASS-21) is a public domain questionnaire intended to measure and differentiate affective symptoms. The original English version, shortened from a 42-item questionnaire (the DASS-42), has been extensively validated (Antony et al., 1998; Chin et al., 2019; Crawford et al., 2011; Crawford & Henry, 2003; Lee et al., 2019; Osman et al., 2012), but the Finnish version has not. The DASS-21 has been noted for its “cleaner” factor structure compared with the 42-item version (Crawford et al., 2011). It has appeared in thousands of original research articles and been translated into more than 40 languages (Lee et al., 2019; Scholten et al., 2017). It has satisfactory measurement invariance across languages and countries, but mean scores vary (Bibi et al., 2020; Scholten et al., 2017; Zanon et al., 2021). The variation may stem from differences in translation and culture, so language-specific norms are sorely needed. Previous Finnish research has relied on data from English-speaking populations (Räsänen et al., 2016; Sourander & Marttunen, 2016).

In the DASS-21, symptoms are assumed to correspond to clinical disorders but on a scale that includes subclinical variation (Lovibond & Lovibond, 1995). Some of the items are derived directly from the first edition of the Beck Depression Inventory-I (BDI-I) and the Beck Anxiety Inventory (BAI), but the test also contains a third factor, Stress (Lovibond & Lovibond, 1995; see also Beck et al., 1988). The stress subscale should not be confused with the biomedical concept of stress, which refers to an external stressor. Rather, the Stress subscale concerns symptoms of affective distress with questions such as “I found myself getting agitated” or “I felt that I was rather touchy.” By contrast, the Anxiety subscale concerns physical, panic-related symptoms, such as trembling, elevated heart rate, and dryness of the mouth. For the English original and the Finnish translation, see Supplemental Material 1.

While the DASS-21 is a robust and widely applicable survey instrument, its structure is unclear. The DASS-21 was designed to measure three types of symptoms (Lovibond & Lovibond, 1995). In practice, the number of factors in best fitting models ranges between one and three (Antony et al., 1998; Bibi et al., 2020; Daza et al., 2002; Duffy et al., 2005; Lee et al., 2019; Lovibond & Lovibond, 1995; Tran et al., 2013). In a systematic review, Lee et al. (2019) argued that solutions other than the three-factor solution are supported by “inadequate and doubtful” evidence. Still, there is notable intercorrelation between the subscales ($r = .46-.72$; Antony et al., 1998). The overlap is accounted for by a bifactorial structure first proposed for the original DASS-42 (Crawford & Henry, 2003; see also Shankman & Klein, 2003). This model adds a fourth factor called General Distress (GD), which has been successfully applied to the DASS-21 among adults (Chin et al., 2019; Lee et al., 2019; Osman et al., 2012; Zanon et al., 2021) and adolescents (Tully et al., 2009).

We administered the DASS-21 to two samples of Finnish-speaking municipal workers from a variety of socioeconomic backgrounds and ages (24–82). The samples make up a broad cross-section of a country’s adult population. To our knowledge, it is the largest used to evaluate the DASS-21 in any language so far (Lee et al., 2019), which allows us to investigate the differential item functioning by gender, age, and response language. For the first time, DASS-21’s invariance over age was studied in the whole interval from young adults to retirees.

We examined the validity and reliability of using the DASS-21 to measure affective symptoms among Finnish speakers. We investigated the measurement error, which remains understudied despite its importance in interpreting reliability (Lee et al., 2019). As DASS-21 scores’ distributions are known to vary between countries and languages (Scholten et al., 2017), we investigated the invariance between genders and age groups under the unifactorial model (Figure 1a). We further compared the distribution properties between Finnish responses and non-Finnish responses. Finally, we produced normative data and statistics for Finnish speakers.

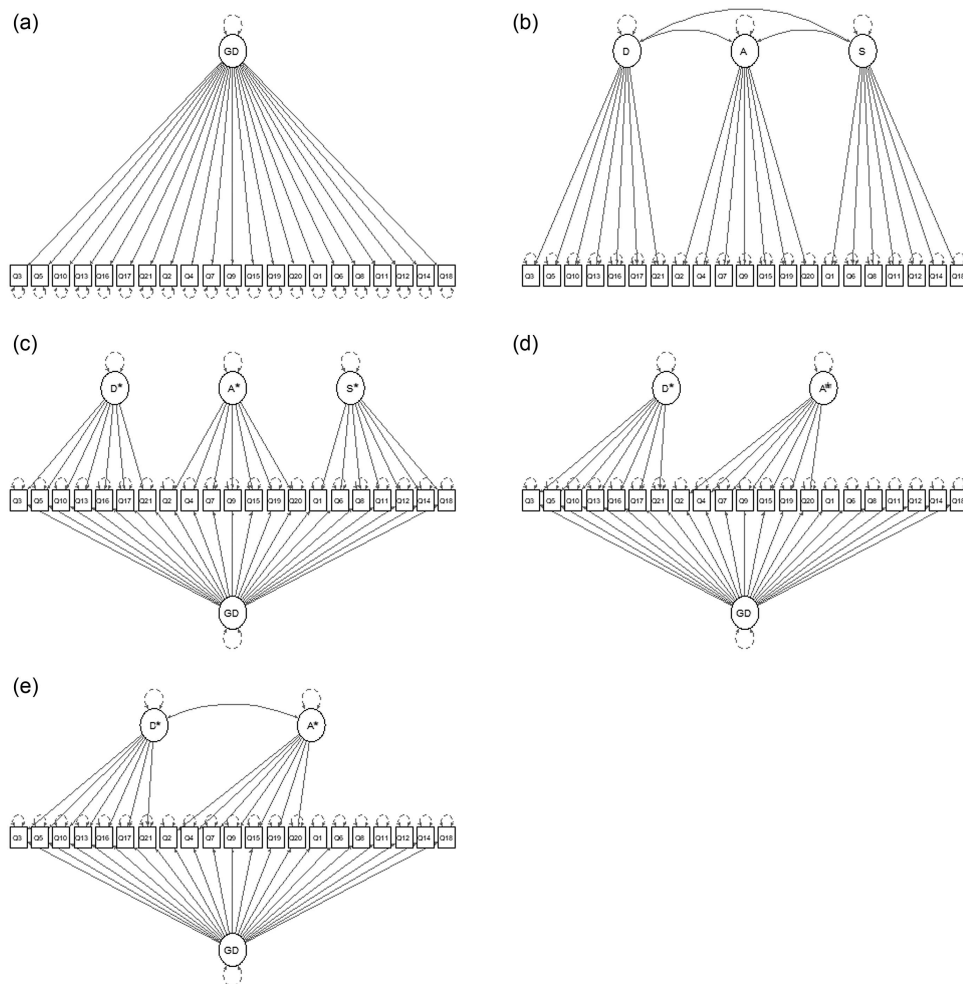
Method

Participants and Data Collection

This study is a part of the Helsinki Health Study (Lahelma et al., 2013; Lallukka et al., 2020), an ongoing longitudinal study of employee health in Finland. The data come from two cohorts of present or former employees of the city of Helsinki. One consists of working-age adults (24–45 years old in 2022; Lallukka et al., 2020) and the other of older adults (60–80 years old in 2022; see Lahelma et al., 2013). The staff covers hundreds of occupations, including manual workers, routine nonmanuals, professionals, and managers (Lahelma et al., 2013).

The phase 1 survey data among older employees were collected in 2000, 2001, and 2002 when the participants were 40-, 45-, 50-, 55- or 60-year-old employees of the city of Helsinki. They have been followed up four times: in 2007, 2012, 2017, and 2022, irrespective of their work status at each follow-up. The older cohort participated by mail only (Lahelma et al., 2013; Lallukka et al., 2020). There was an additional telephone survey to catch nonrespondents in 2022. However, the short telephone survey did not include the DASS-21, and therefore telephone participants ($n = 285$) could not be included in the present study. The second cohort (currently working age) was initiated in 2017 among all employees of the city of Helsinki under the age of 40, whose employment had lasted for at least 4 months and who were employed at least 50% of the time. They were followed up in 2022. Of them, 2,430 answered online and 776 by mail. Only the 2022 phase of each cohort included the DASS-21. We assume that the scores remain invariant over the mode of administration.

The final combined sample size was 8,563 with 6,961 women. The sample formation is detailed in Figure 2. Sociodemographic information of the two cohorts is presented in Table 1. The participants were able to take the surveys in Finnish, Swedish, English, or Russian (in the working-age cohort) or Finnish or Swedish (in the older cohort). We only included those answering in Finnish for this study (378 excluded; see Figure 2). The Finnish-language sample did not meaningfully differ from the sample who answered the questionnaire in a language other than Finnish

Figure 1*The Five Structural Models for the Depression Anxiety Stress Scales*

Note. GD = general distress; D = depression; A = anxiety; S = stress; D* = residual D factor; A* = residual A factor; S* = residual S factor; Q = question.

(Supplemental Material 2). We excluded all participants with missing values for the Emotional Well-being (EW) subscale of the RAND-36 (named after the RAND Corporation, an American nonprofit).

Nonparticipation analyses have shown the original cohorts to be broadly representative of the target populations, though younger employees and men are slightly underrepresented, as are those with lower income, those in manual jobs, and those with more sickness absence (Lahelma et al., 2013; Lallukka et al., 2020). We further investigated the attrition between the 2017 and 2022 measurements (Supplemental Material 3). In both cohorts, a higher level of education was associated with attrition. In the older cohort, lower health-related quality of life (RAND-36) in 2017 predicted attrition in 2022. The effect was not found in the working-age cohort. Age had a statistically significant but weak and inconsistent effect on attrition: In the working-age cohort, older age was associated with attrition. In the older cohort, it was younger age. Marital status and employment class were not associated with attrition.

Ethics

Permits for data collection were obtained from the city of Helsinki. The plan for the present study was approved by the ethics committee of the University of Helsinki (no protocol number). The study was not preregistered.

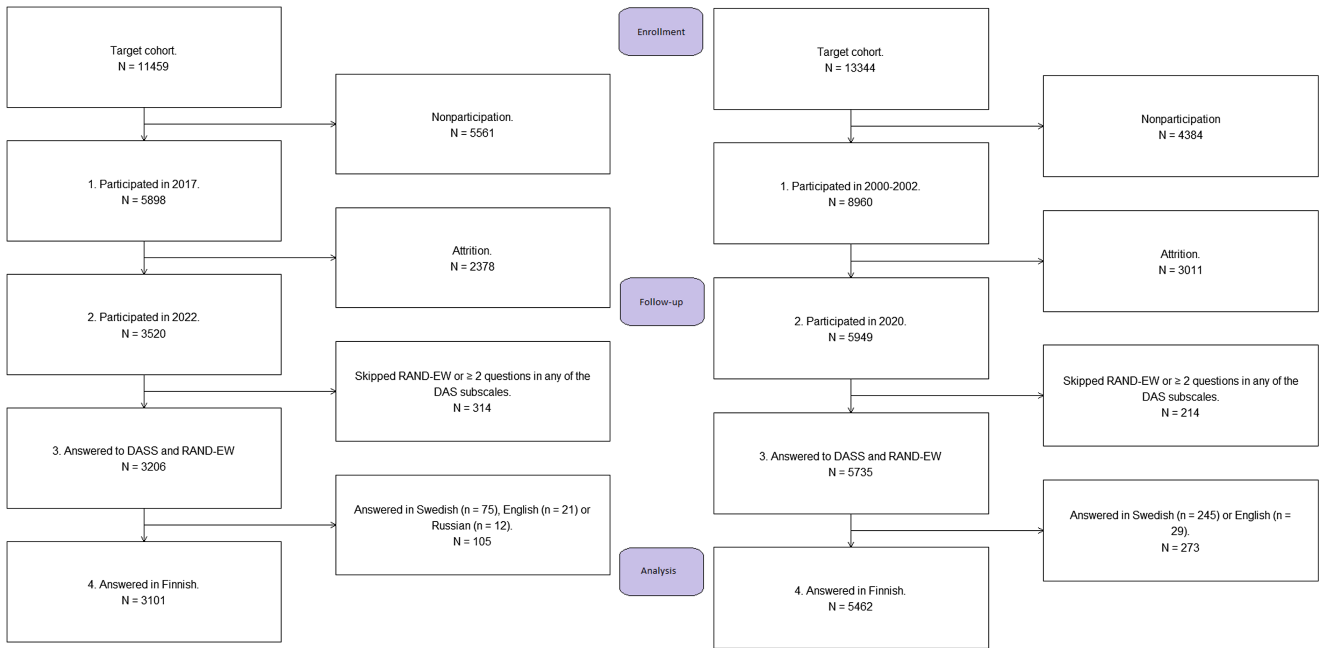
Materials

DASS-21

The DASS-21 consists of 21 statements (e.g., “I felt that life was meaningless,” “I found it hard to wind down,” and “I was aware of dryness of my mouth”; see Supplemental Material 1). There are seven items to each of the three subscales: Depression, Anxiety, and Stress (Antony et al., 1998; Lovibond & Lovibond, 1995). Participants were instructed to rate the statements on a 4-point Likert scale based on how well the statements applied to them over the last week. The answers were coded 0–3 for a maximum of 21 points.

Figure 2

The Sample Formation in the Two Helsinki Health Study Cohorts: The Working Age (Left) and the Older (Right)



Note. DASS = Depression Anxiety Stress Scales; RAND-EW = RAND Emotional Well-being. See the online article for the color version of this figure.

The GD factor was measured using all 21 items in the questionnaire for a maximum of 63 points (Henry & Crawford, 2005; Osman et al., 2012).

All participants missing more than one question for any one of the DASS subscales were excluded. Missing values were imputed with the mean value of the other six items in the subscale, as suggested by the original author Lovibond (2022, item 27). In the working-age cohort ($N = 3,206$), DASS values were imputed for 184 participants (5.7%; $n_d = 76$, $n_a = 66$, $n_s = 42$). In the older cohort ($N = 5,736$),

they were imputed for 271 participants (4.7%; $n_d = 99$, $n_a = 83$, $n_s = 89$).

The original Finnish DASS-21 adaptation was developed by Raimo Lappalainen of the University of Jyväskylä, Finland. The initial translation was reviewed by an expert board of three researchers, all native Finnish speakers, and the three subscales were found internally consistent in cursory analyses (Räsänen et al., 2016). The translation was revised for the present study. The updated version is available in Supplemental Material 1 and on the

Table 1

Sociodemographic Characteristics in 2022

Measure	Working-age cohort (ages 24–45)			Older cohort (ages 60–82)		
	Men	Women	Overall	Men	Women	Overall
Sample size	613	2,488	3,101	989	4,473	5,462
Age						
<i>M</i> (<i>SD</i>)	37.8 (4.43)	37.2 (4.59)	37.3 (4.56)	71.2 (6.55)	70.0 (6.40)	70.2 (6.44)
<i>Mdn</i>	38	38	38	71	70	71
Missing	0 (0%)	0 (0%)	0 (0%)	84 (8.5%)	287 (6.4%)	371 (6.8%)
Highest educational attainment, ISCI (United Nations Educational, Scientific and Cultural Organization Institute for Statistics, 2012)						
Middle school or below (1–2)	9 (1.5%)	12 (0.5%)	21 (0.7%)	169 (17.1%)	735 (16.4%)	904 (16.6%)
High school or equivalent (3–4)	252 (41.1%)	693 (27.9%)	945 (30.5%)	457 (46.2%)	2,480 (55.4%)	2,937 (53.8%)
University or equivalent (5–8)	349 (56.9%)	1,777 (71.4%)	2,126 (68.6%)	358 (36.2%)	1,216 (27.2%)	1,574 (28.8%)
Missing	3 (0.5%)	6 (0.2%)	9 (0.3%)	5 (0.5%)	42 (0.9%)	47 (0.9%)
Marital status						
Single	162 (26.4%)	744 (29.9%)	906 (29.2%)	80 (8.1%)	510 (11.4%)	590 (10.8%)
Married or cohabiting	429 (70.0%)	1,604 (64.5%)	2,033 (65.6%)	705 (71.3%)	2,487 (55.6%)	3,192 (58.4%)
Divorced or widowed	19 (3.1%)	132 (5.3%)	151 (4.9%)	173 (17.5%)	1,389 (31.1%)	1,562 (28.6%)
Missing	3 (0.5%)	8 (0.3%)	11 (0.4%)	31 (3.1%)	87 (1.9%)	118 (2.2%)

Note. ISCI = International Standard Classification of Education.

home page of the University of New South Wales at <https://www2.psy.unsw.edu.au/groups/dass/Finnish/Finnish.htm>.

In addition to the original three-factor model of the DASS-21, we tested several other structural models (Figure 1), including a one-factor model (Model a), a bifactor model (Model c), and two bifactorlike models (Models d and e). Models c–e contain a general factor called GD, which significantly affects the interpretation of the results (Chin et al., 2019; Crawford & Henry, 2003; Lee et al., 2019; Osman et al., 2012; Shankman & Klein, 2003; Tully et al., 2009; Zanon et al., 2021). In these models, the three factors besides GD represent the shared variance after partialing out the GD factor. As such, we term these residual factors D^* , A^* , and S^* .

The RAND-36 Measure of Health-Related Quality of Life

The RAND-36 is a widely used questionnaire for health-related functioning and quality of life (Hays & Morales, 2001). It comprises 36 items that assess eight subscales (Hays & Morales, 2001). We used the EW subscale for our main analyses. EW measures subjective emotional experience (Hays & Morales, 2001) or moods (Aalto et al., 1999) by five items, which contain words such as “nervous,” “happy,” and “downhearted.” Each item is scored on a 6-point Likert scale ranging from *all of the time* to *none of the time*. The Finnish version of the questionnaire exhibits good structural validity and internal consistency (Aalto et al., 1999). The EW subscale acted as a proxy for symptoms of affective mental health, meaning we expected it to correlate negatively with all DASS-21 subscales. All eight subscales were used for attrition analyses. Beside EW, there are Physical Functioning, Physical Role Functioning, Emotional Role Functioning, Energy/Fatigue, Emotional, Social Functioning, Pain, and General Health.

Validity

The *Standards* define validity as “the degree to which evidence and theory support the interpretations of test scores for proposed uses of tests” (American Educational Research Association [AERA] et al., 2014). We evaluated the use of the DASS-21 to measure affective symptoms among Finnish speakers in research and clinical practice. We report validity evidence in five categories: (1) test content, (2) response process, (3) internal structure, (4) relations to other variables, and (5) the consequences testing (AERA et al., 2014; Sireci & Benítez, 2023).

1. Test content is relevant insofar as it relates to the measured object (AERA et al., 2014). Test content should be both relevant and generalizable to the target population (Hernández et al., 2020). Translators must also ensure that “what is assessed should be understood in the same way across language and cultural groups” (International Test Commission [ITC], 2017).
2. Response processes provide evidence for validity when they overlap with the intended construct (AERA et al., 2014). In particular, the target population should be familiar with the conventions of the test. In translation, it is important that the populations be similar in method of administration, time restrictions, and social situation (Hernández et al., 2020; ITC, 2017). For instance, our data come from the multilingual staff of the city of

Helsinki (though the vast majority speak Finnish, and Finnish is the working language for most), and they were collected through a standardized submitted to all employees in the target population to ensure fairness. The non-Finnish answers were used to investigate the effect of response language within the sample.

3. Internal structure refers to the relationships between test items, such as dimensionality, item-specific functioning, and differential item functioning—that is, whether groups differ despite similar status for the relevant construct (AERA et al., 2014). We investigated the dimensionality using parallel analysis and confirmatory factor analysis (CFA) and looked for evidence of divergence between the theorized subscales: Depression, Anxiety, and Stress. Item-specific phenomena were not studied. Differential item functioning was investigated through intergender and interage group invariance analyses.
4. Relations to other variables may be related to variables that the test should predict (criterion validity) or categories expected to differ. The variables may also be ones that the test should be associated with (convergence) or that it should not be associated with (divergence; AERA et al., 2014). We looked for evidence of convergent evidence by investigating relation between the DASS-21 and EW as measured by the RAND-36 (Aalto et al., 1999). We expected them to correlate negatively. We did not investigate the predictive power of the DASS-21 among Finnish speakers.
5. The consequences of testing may be intended or unintended. The DASS-21 was developed to cover and discriminate between depressive and anxiety symptoms as reported by the BDI and the BAI, respectively (Lovibond & Lovibond, 1995). Similarly, we trust that a comprehensive but brief questionnaire would make testing faster and the results more informative compared with longer, more specialized instruments. The DASS-21 is only half the length of the BDI and BAI combined (42 items) and does allow researchers to assess a wide spectrum of affective symptoms. Its clinical use, however, was not assessed quantitatively.

Reliability and Item Statistics

For reliability, we report McDonald’s ω_h and ω_{tot} . The latter may be used to estimate unidimensional reliability, while ω_h is recommended for multidimensional cases (Zinbarg et al., 2005). Therefore, we use ω_h to estimate the reliability of our measurement of the sum score of the test (which represents the GD factor) and ω_{tot} for the three subscales of DASS (Depression, Anxiety, and Stress). Cronbach’s α is given for comparison’s sake. The α is only a special case of the ω s assuming τ -equivalence (Tavakol & Dennick, 2011; Zinbarg et al., 2005), that is, when all items exhibit perfect intercorrelation and identical true-score variance (de Gruijter & van der Kamp, 2019). In psychological measurement, τ -equivalence is rarely a tenable assumption, making Cronbach’s α a lower bound (and a biased estimate) of reliability (de Gruijter & van der Kamp, 2019; Tavakol & Dennick, 2011; Zinbarg et al., 2005).

As reliability coefficients are sample dependent, we present them with the standard error of measurement (SEm , σ_E), which quantifies the effect of error introduced in measurement (de Gruijter & van der Kamp, 2019).

Software

All R code for data transformations and analyses is available at <https://osf.io/6q8fk/> (Vähäsarja, 2024). The participant data are not available due to Finnish data protection legislation.

All statistical analyses were conducted in R 4.2.1 on Windows 10 (R Core Team, 2022). The factor analyses and fit indices were produced using the lavaan package (Rosseel, 2012) for R. Two packages, lavaan and semTools (Jorgensen et al., 2022) were used for the invariance analysis. The Psych package (Revelle, 2023) was used for the investigation of multinormality, kurtosis, skewness, and reliability. The nFactors package (Raiche & Magis, 2022) was used to investigate the number of factors. The WebPower package (Zhang & Mai, 2023) was used for power analyses. All tables were produced using the rempsyc package (Thériault, 2023).

Data Analysis

The data's heavy left-tailed skew is evident in Table 2. Therefore, we opted for robust estimation methods, which are detailed in their respective analyses.

In the factor analyses, we used the polychoric correlation matrices provided in Supplemental Material 4 (Olsson, 1979; Rosseel, 2012)

with the diagonally weighted-least-squares method of parameter estimation, which has been shown to be an unbiased estimator for ordinal data (Li, 2016). Though it assumes a normal distribution for each latent variable, no such assumption is made for observed variables (Li, 2016). The number of factors was assessed using parallel analysis with 100 iterations (Thériault, 2023). We estimated the fits for the five structural models derived from the theory (Figure 1). We used the chi-square statistic and several additional indices with the following cutoffs, based on statistical theory and previous validations of the DASS-21 in other languages: For the comparative fit index and the Tucker–Lewis index, we used .95 (Ahmed et al., 2022; Hu & Bentler, 1999; van de Schoot et al., 2012). For the root-mean-square error of approximation (RMSEA), we used a cutoff of .06 (Ahmed et al., 2022; Henry & Crawford, 2005; Hu & Bentler, 1999; Tully et al., 2009), and for the standardized root-mean-square residual, we set it at .08 (Ahmed et al., 2022; Henry & Crawford, 2005; Tully et al., 2009).

We tested for intergender and interage group invariance following van de Schoot et al.'s (2012) checklist. First, we fitted the configural model. That is, we compared the fit of the one-factor model (Plot a in Figure 1) through multiple CFAs, one for men and one for women, in each cohort. Then we progressively constrained the model. First, we tested metric invariance by constraining the factor loadings across groups as equal but allowing intercepts to vary. Then, scalar invariance was tested by constraining both loading and intercepts as equal across groups. Finally, we tested the strict invariance model, which fixes not only factor loadings and intercepts but also error variances. We applied Hu and Bentler's (1999) cutoffs for the model

Table 2
Sample Distribution and Reliability Statistics in 2022

Item	Range	Working-age cohort (ages 24–45; $N = 3,101$)									Older cohort (ages 60–82; $N = 5,462$)								
		M	Mdn	SD	Skew	Kurt	α	ω_h	ω_{tot}	SEm	M	Mdn	SD	Skew	Kurt	α	ω_h	ω_{tot}	SEm
D	0–21	3.48	2	4.00	1.81	3.39	.91	.83	.92	1.13	3.28	2	3.71	1.81	3.78	.89	.78	.90	1.17
A	0–21	2.39	1	2.88	2.08	5.60	.78	.65	.80	1.29	2.42	2	2.73	2.16	6.70	.76	.60	.78	1.28
S	0–21	5.24	5	4.05	1.02	0.92	.88	.73	.89	1.34	3.56	3	3.35	1.36	2.30	.85	.73	.87	1.21
GD	0–63	11.1	8	9.66	1.55	2.72	.93	.75	.94	4.83	9.26	7	8.64	1.82	4.51	.92	.80	.93	3.87
Q1	0–7	0.90	1	0.85	0.70	–0.13					0.49	0	0.68	1.35	1.68				
Q2	0–7	0.67	0	0.81	1.07	0.49					0.83	1	0.86	0.85	0.03				
Q3	0–7	0.43	0	0.69	1.64	2.28					0.38	0	0.64	1.75	2.84				
Q4	0–7	0.27	0	0.60	2.35	5.35					0.27	0	0.60	2.52	6.45				
Q5	0–7	0.97	1	0.82	0.63	–0.02					0.80	1	0.73	0.79	0.64				
Q6	0–7	0.80	1	0.83	0.77	–0.11					0.57	0	0.71	1.14	0.97				
Q7	0–7	0.18	0	0.51	3.19	10.85					0.23	0	0.61	2.94	8.71				
Q8	0–7	0.57	0	0.74	1.18	0.90					0.35	0	0.60	1.83	3.52				
Q9	0–7	0.40	0	0.71	1.86	2.97					0.23	0	0.55	2.72	8.02				
Q10	0–7	0.34	0	0.70	2.17	4.27					0.42	0	0.72	1.79	2.79				
Q11	0–7	1.13	1	0.75	0.41	0.04					0.81	1	0.67	0.59	0.69				
Q12	0–7	0.94	1	0.81	0.64	–0.01					0.59	0	0.71	1.11	1.04				
Q13	0–7	0.69	1	0.78	0.99	0.48					0.62	0	0.73	1.08	0.95				
Q14	0–7	0.35	0	0.64	1.91	3.38					0.30	0	0.58	2.14	4.88				
Q15	0–7	0.19	0	0.51	3.02	9.79					0.12	0	0.43	4.14	19.29				
Q16	0–7	0.38	0	0.68	1.86	3.10					0.41	0	0.66	1.63	2.49				
Q17	0–7	0.40	0	0.73	1.91	3.07					0.37	0	0.68	1.98	3.66				
Q18	0–7	0.53	0	0.72	1.26	1.11					0.44	0	0.65	1.45	1.98				
Q19	0–7	0.40	0	0.70	1.79	2.67					0.46	0	0.74	1.66	2.36				
Q20	0–7	0.26	0	0.59	2.45	5.86					0.27	0	0.57	2.31	5.71				
Q21	0–7	0.26	0	0.62	2.66	6.98					0.27	0	0.61	2.57	6.70				

Note. Standard error of measurement was defined as $SEm = SD \times \sqrt{1 - \text{reliability}}$. Reliability was estimated with the coefficient ω_{tot} for the subscales D, A, and S, while ω_h was used for GD, per Zinbarg et al. (2005). Skew was estimated as G_1 and kurtosis as G_2 . For details, see Joanes and Gill (1998). SEm = standard error of measurement; D = Depression; A = Anxiety; S = Stress; GD = General Distress.

fit indices, as described in the paragraph above. Latent means between groups were assessed using Welch's unequal variance t tests with Bonferroni correction.

Results

Validity Evidence From Test Content

The DASS-21 was translated into Finnish by a team of three native Finnish-speaking experts knowledgeable in affective disorders and testing for them as well as Finnish culture. This accords with the ITC's (2017) recommendations. No outside translators or piloting was involved. The concepts of affective symptoms were deemed clear and relevant to the target population. The effect of response language on the distribution properties was investigated to assess generalizability and fairness (AERA et al., 2014), and no meaningful discrepancy was detected.

Validity Evidence From Response Processes

The translation team deemed the language and format of the translation comprehensible to the target population, as cautioned by the ITC (2017) and Hernández et al. (2020): The DASS-21 is made up of statements on a Likert scale, as are other instruments used among Finnish adults, such as the RAND-36, which this study's target population is previously familiar with (Aalto et al., 1999; Lahelma et al., 2013; Lallukka et al., 2020). The data were collected in a standardized fashion among all employees of the city of Helsinki in the target cohorts, with similar time restrictions but varying in the mode of administration: The older cohort and a minority of the working-age cohort answered by mail, while the rest answered online. The participants could answer not just in Finnish but also in Swedish, English, or Russian (though only Finnish or Swedish in the older cohort). We compared the distribution properties between Finnish and non-Finnish answers and observed no discrepancy (Supplemental Material 2), which indicates that the concepts were not language or culture specific.

Validity Evidence From Internal Structure

Parallel analysis ($n_{\text{iterations}} = 100$) suggested a five-factor solution in both cohorts (Figure 3). However, parallel analysis is known to overfit with a higher sample size (Revelle, 2023). The subjective "elbow" after factor 1 is apparent in both cohorts—that is, the first eigenvalue is considerably higher with the curve quickly flattening after the second datapoint (working age: $\lambda_1 = 11.01$, $\lambda_2 = 1.00$, $\lambda_3 = 0.70$; older: $\lambda_1 = 10.63$, $\lambda_2 = 0.82$, $\lambda_3 = 0.58$). This suggests that most of the variance in the data is explained by the first factor.

The five models depicted in Figure 1 were compared using CFA. The cutoffs for the four indices (comparative fit index, Tucker–Lewis index, RMSEA, and standardized root-mean-square residual) allowed for all models in both cohorts (Table 3), with the single exception of RMSEA in the working-age cohort ($.08 > .06$, the cutoff). As all models fit well enough, the parsimonious one-factor model should be preferred. The three theorized factors' intercorrelations (Model b; see Figure 1) were high both among the working-age cohort ($R_{DA} = .75$, $R_{DS} = .82$, $R_{AS} = .79$) and the older one ($R_{DA} = .76$, $R_{DS} = .85$, $R_{AS} = .83$), and in both cohorts, 20/21

items loaded more heavily onto the GD factor than their respective subscale (Table 4). After fitting the four-factor bifactorial solution, 417 out of the 420 item covariances in the two cohorts were above $-.05$ and below $.05$ (Supplemental Material 4), which suggested no further relevant factors.

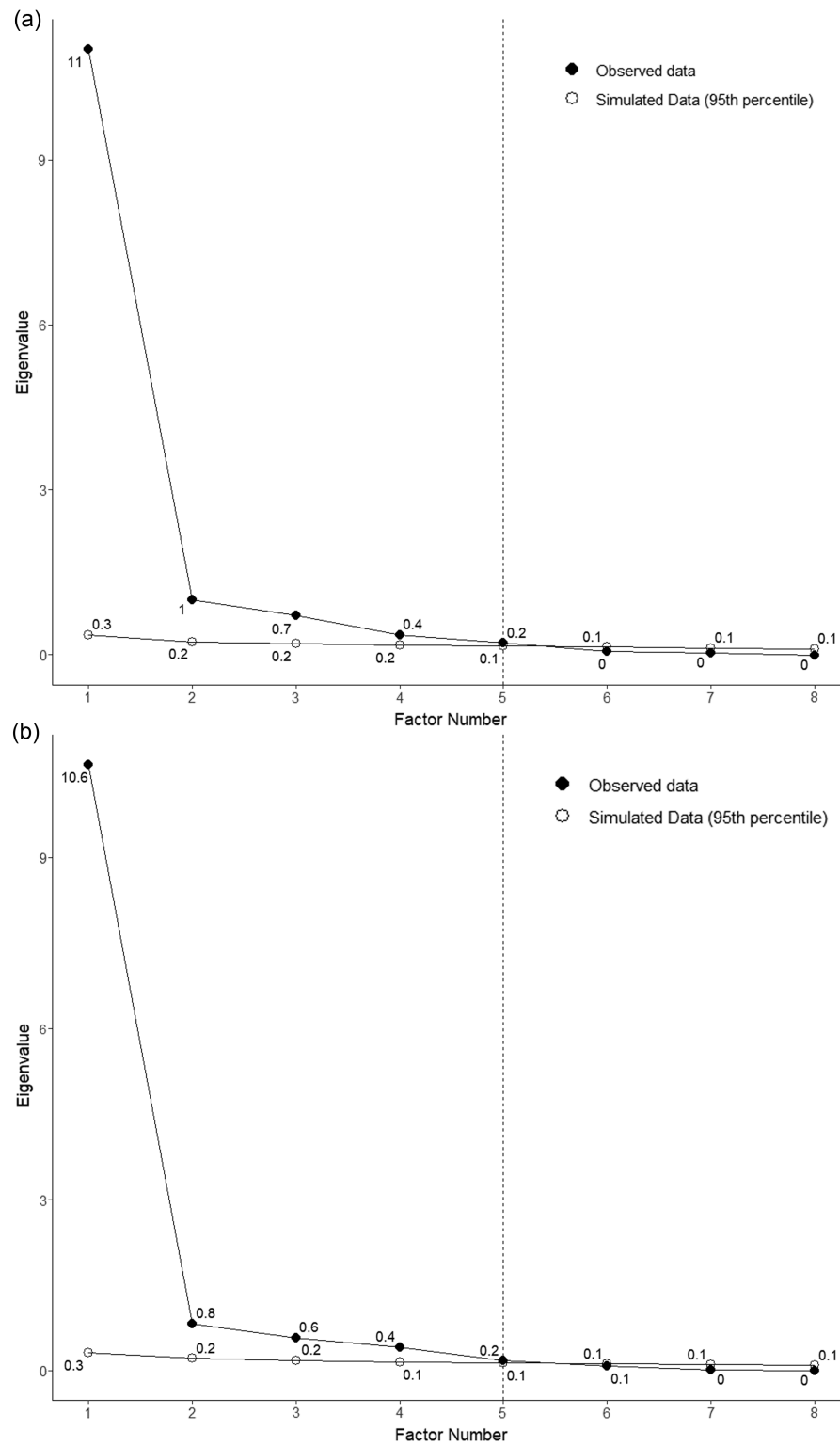
Validity Evidence From Relations to Other Variables

For evidence of convergent validity, we compared the participants' DASS-21 scores with the previously validated RAND-36 EW subscale (Aalto et al., 1999). All three subscales correlated negatively with the RAND-36 EW subscale both in the working-age cohort ($R_D = -.77$, $R_A = -.52$, $R_S = -.69$, $R_{GD} = -.76$) and the older one ($R_D = -.73$, $R_A = -.48$, $R_S = -.67$, $R_{GD} = -.72$), suggesting that the DASS-21 did indeed measure poor mental well-being in the two cohorts.

No differential item functioning based on age or gender was observed. Within cohort, age-related effects were observed neither in the working-age cohort ($R_{D\text{-age}} = -.02$, $R_{A\text{-age}} = .02$, $R_{S\text{-age}} = -.04$, $R_{GD\text{-age}} = -.08$) nor the older one ($R_{D\text{-age}} = .03$, $R_{A\text{-age}} = .03$, $R_{S\text{-age}} = -.01$, $R_{GD\text{-age}} = .05$). No difference in distribution or reliability was observed between Finnish answers and the whole sample (Supplemental Material 2). The parsimonious unifactorial model (Plot a in Figure 1) was deemed to best fit the data, so it was used to investigate intergender and interage group measurement invariance. According to van de Schoot et al.'s (2012) workflow, all models showed a good fit (Tables 5 and 6). The good fit of the strict model implies that the DASS-21 measures the latent psychopathological constructs identically across Finnish-speaking men and women as well as among working-age and older adults (Hu & Bentler, 1999; van de Schoot et al., 2012), and so latent means and errors between the four groups may be compared unadjusted (van de Schoot et al., 2012). Further investigation of men's and women's mean scores revealed a significant difference in the Stress subscale in the working-age cohort. Even then, the effect was <1 on a scale of 0–21. The effect of age on DASS-21 scores appears to be minor but worth considering: The younger of our two cohorts reported slightly higher symptoms of depression and stress, but no effect was found within cohort. The discrepancy is, nevertheless, accounted for by separate norms for those under 45 and those above (Supplemental Material 5).

Reliability

Psychometric statistics are reported in Table 2. The scores from all three theorized subscales appeared reliable enough for research. The reliability coefficients for the scores from each subscale but Anxiety exceeded $.80$, which supports future use in individual assessment among Finnish-speaking adults. Note that we used ω_{tot} for the three original subscales and ω_h for the GD factor. Anxiety, too, exceeded $.75$ in both cohorts. Standard errors of measurement (SEm) ranged between 1 and 1.5 for Depression, Anxiety, and Stress. The SEm s were between 3.5 and 5 for the GD factor. Converting the SEm s to .95 confidence intervals yields ± 2 for all three subscales: Depression, Anxiety, and Stress ($\pm 1.96 \times SEm$; de Gruiter & van der Kamp, 2019). For GD, the .95 confidence interval is ± 9 in the working-age cohort and ± 8 in the older one.

Figure 3*Parallel Analysis in the Two Cohorts: Working Age (a; Ages 24–45) and Older (b; Ages 60–82)*

Note. $N_{\text{iterations}} = 100$. The dashed line is located at the suggested solution of five factors.

Table 3*Model Fit Indices From Confirmatory Factor Analyses on the Depression Anxiety Stress Scales*

Model	Working-age cohort (ages 24–45; <i>N</i> = 3,101)						Older cohort (ages 60–82; <i>N</i> = 5,462)					
	χ^2 (df)	χ^2/df	CFI	TLI	RMSEA	SRMR	χ^2 (df)	χ^2/df	CFI	TLI	RMSEA	SRMR
a	3,954 (189)	20.92	0.98	0.98	0.08	0.08	4,243 (189)	22.45	0.99	0.98	0.06	0.06
b	1,738 (186)	9.34	0.99	0.99	0.05	0.05	1,922 (186)	10.33	0.99	0.99	0.04	0.04
c	835 (168)	4.97	1.00	1.00	0.04	0.04	1,177 (168)	7.00	1.00	1.00	0.03	0.03
d	1,302 (175)	7.44	1.00	0.99	0.05	0.04	1,483 (175)	8.48	1.00	0.99	0.04	0.04
e	1,164 (174)	6.69	1.00	0.99	0.04	0.04	1,385 (174)	7.96	1.00	1.00	0.04	0.04
Cutoffs (null)		<2	≥.95	≥.95	≤.06	≤.08		<2	≥.95	≥.95	≤.06	≤.08

Note. For the models, see Figure 2. Fits are estimated using the robust weighted-least-squares method. Model a is unifactorial, Model b has three intercorrelated factors, Model c is bifactorial with three uncorrelated factors and a fourth general factor, Model d is the same as c but with one less factor, and Model e is the same as d but with intercorrelated factors (Figure 2). CFI = comparative fit index; TLI = Tucker–Lewis index; RMSEA = root-mean-square error of approximation; SRMR = standardized root-mean-square residual.

Prevalence of Affective Symptoms

Both cohorts exhibited low mean rates of Depression, Anxiety, and Stress, with mean values ranging between 3 and 5.5 out of a maximum of 21 per subscale (Table 2). Consequently, rates of GD were low. The single outlier was the rate of reported stress in the working-age cohort, which was high for both genders compared with the other subscales and the older cohort (Table 7). An intergender comparison showed that women in the working-age cohort showed significantly more stress symptoms than men ($p < .001$; Table 7). The mean difference was .8 on a scale of 0–21.

Discussion

The DASS-21 is a robust measure of affective symptoms among Finnish-speaking working population and retirees. The content is

clear and relevant to the target population. No differential item functioning by age or gender was observed. As previously reported, the translation exhibits adequate reliability to measure Depression, Anxiety, and Stress in questionnaire-based studies on Finnish adults (Räsänen et al., 2016). As for individual assessment among Finnish-speaking adults, scores from all subscales but Anxiety appear sufficiently reliable. The use of DASS-21 and the estimation of its measurement error are discussed below.

However, little evidence of divergence between the factors was found. A unifactorial model fit the data well, and all subscales were clearly (and negatively) associated with the EW subscale of the RAND-36, which has been previously validated in Finnish (Aalto et al., 1999). In a three-factor solution (Figure 1, Model b), the intercorrelations ran high (.61–.73), and, in a bifactor solution (Model c), 20 out of 21 items loaded more heavily onto the added GD factor than their respective residual factors, in both cohorts.

Table 4*Estimates for the Standardized Factor Loadings From the Bifactorial Solution to the Confirmatory Factor Analysis*

Item	Working-age cohort (ages 24–45; <i>N</i> = 3,101)								Older cohort (ages 60–82; <i>N</i> = 5,462)							
	GD		D		A		S		GD		D		A		S	
	Slope	SE	Slope	SE	Slope	SE	Slope	SE	Slope	SE	Slope	SE	Slope	SE	Slope	SE
Q3	.49	.02	.23	.02					.42	.01	.17	.01				
Q5	.50	.02	.05	.02					.40	.01	.15	.01				
Q10	.42	.02	.40	.02					.44	.01	.39	.01				
Q13	.57	.02	.23	.02					.50	.01	.20	.01				
Q16	.45	.02	.32	.02					.45	.01	.27	.01				
Q17	.47	.02	.36	.02					.43	.01	.25	.01				
Q21	.35	.02	.38	.01					.37	.01	.32	.01				
Q2	.24	.02			.15	.02			.28	.01			.20	.02		
Q4	.25	.02			.26	.02			.23	.01			.28	.02		
Q7	.20	.01			.18	.02			.23	.01			.18	.02		
Q9	.45	.02			.10	.02			.34	.01			.06	.01		
Q15	.30	.02			.16	.02			.26	.01			.07	.01		
Q19	.31	.02			.33	.02			.30	.01			.28	.02		
Q20	.34	.02			.18	.02			.35	.01			.11	.01		
Q1	.54	.02					.42	.02	.43	.01					.31	.02
Q6	.55	.01					.11	.02	.44	.01					.09	.02
Q8	.55	.01					.19	.02	.44	.01					.14	.01
Q11	.48	.01					.16	.02	.42	.01					.05	.02
Q12	.58	.01					.37	.02	.52	.01					.20	.02
Q14	.42	.02					–.03	.02	.36	.01					–.07	.02
Q18	.49	.02					–.03	.02	.41	.01					–.06	.02

Note. Values above .40 are presented in bold. Data collected in 2022. CFA = confirmatory factor analysis; GD = General Distress; D = Depression; A = Anxiety; S = Stress; SE = standard error; Q = question.

Table 5*Intergender Invariance Analyses Between Men and Women*

Model	Working-age cohort (ages 24–45; <i>N</i> = 3,101)				Older cohort (ages 60–82; <i>N</i> = 5,462)			
	CFI	TLI	RMSEA	SRMR	CFI	TLI	RMSEA	SRMR
Configural	.978	.976	.043	.063	.982	.980	.033	.051
Metric	.975	.974	.044	.067	.981	.980	.033	.054
Scalar	.972	.971	.046	.069	.980	.980	.034	.054
Strict	.970	.971	.046	.071	.979	.980	.033	.057
Cutoffs (null)	>.950	>.950	<.060	<.080	>.950	>.950	<.060	<.080

Note. Fits are estimated using the robust weighted-least-squares method estimation method. The model is unifactorial (a, Figure 2). Data collected in 2022. CFI = comparative fit index; TLI = Tucker–Lewis index; RMSEA = root-mean-square error of approximation; SRMR = standardized root-mean-square residual.

A bifactor model is also more difficult to score. It requires factor scores, while scales from the one-factor and three-factor solutions (Plots a and b in Figure 1) may be scored using linear sums, which is simple and practical. Finally, it is more difficult to interpret the residual factors D^* , A^* , and S^* than the original factors D , A , and S . The residual factors represent variance unrelated to GD, and as such, they do not measure depression, anxiety or stress as the terms are normally understood.

Affective symptoms appeared unifactorial among Finnish adults, as they have before in other non-English-speaking countries (Zanon et al., 2021). The result may reflect cultural differences or the general ambiguity that permeates the diagnosis of depression and anxiety disorders. Depression and anxiety often co-occur and appear to share both genetic and environmental risk factors, leading many to criticize the categorical approach (Caspi et al., 2014; Goodkind et al., 2015; Kotov et al., 2017; Newman et al., 1998; Rosenström et al., 2019; Shankman & Klein, 2003; Wang et al., 2017). While all subscales reflect negative emotionality, their timescales may differ. The Depression subscale appears to measure features of mood, that is, a transient emotional state, with statements such as “I felt that life was meaningless” or “I felt I wasn’t worth much as a person.” Meanwhile, Anxiety and Stress seem to refer to shorter term effects (“I found myself getting agitated” to “I was aware of a dryness in my mouth”). Differentiating momentary sensations may not be possible through questionnaire studies with long timescales. Our results surely speak against it in the context of the Finnish DASS-21. Future users should carefully consider whether they seek to measure poor

mental well-being or specific affective symptoms. The breadth and brevity of the DASS-21 make it a good measure of affect beyond psychiatric nosology, and it offers a way to study depression and anxiety as they are conventionally defined, but it does not appear to differentiate between them in epidemiological data.

Using Finnish data (Supplemental Material 5) is paramount for any Finnish research or clinical application, as DASS-21 scores vary between translations and countries (Scholten et al., 2017). For instance, quartiles for the Depression subscale were approximately 1, 2, and 5 in both Finnish cohorts, compared with 0, 2, and 4 among U.K. respondents (Henry & Crawford, 2005; Supplemental Material 2). As a rule of thumb, a participant’s score on the sum score (which is related to, but not equal to, the GD factor) may be translated to a .95 confidence interval by adding ± 9 among working-age adults and ± 8 among older adults. For the three subscales, that number is ± 2 . For instance, if an individual scores 6/21 on the Depression subscale, the .95 confidence interval is 4–8, which places them between the 70th and 88th percentile.

Constraints on Generality

We captured a broad sample of Finnish adults from different socioeconomic classes, educational backgrounds, and age groups. Still, we studied two populations of urbanites who were employed or previously employed and now retired. We expected the so-called healthy worker effect to deflate the prevalence of psychopathology relative to the whole population (Wilcosky & Wing, 1987) but found reported symptoms to be higher than in a previous study on an English-speaking general population sample. Whether the effect is explained by a difference in translation or culture requires future research. In any case, the importance of using Finnish norms is highlighted.

Furthermore, as the participants were employees or retired former employees of the city of Helsinki, they are representative of the municipal sector, but not the Finnish labor force or general population. For one, wage and salary earnings of municipal employees were, on average, lower than earnings in the central government and the private sector (Statistics Finland, 2023). The sector is also female-dominated. However, the Finnish municipal sector is large and carries responsibilities that exceed those laid on many of its foreign equivalents. In the city of Helsinki, health and social care (45%) and education (33%) are the largest occupational branches, and the most common occupational titles include nurse, class teacher, kindergarten teacher, and care worker (Lahti et al., 2023).

Table 6*Intergender Invariance Analyses Between Working-Age and the Older Cohorts*

Model	CFI	TLI	RMSEA	SRMR
Configural	.9780	.978	.037	.055
Metric	.976	.974	.040	.060
Scalar	.963	.963	.048	.065
Strict	.961	.962	.049	.070
Cutoffs (null)	>.950	>.950	<.060	<.080

Note. Fits are estimated using the robust weighted-least-squares method estimation method. The model is unifactorial (a, Figure 2). The working-age cohort (*N* = 3,101) included ages 24–45. The older cohort (*N* = 5,462) included ages 60–82. Data collected in 2022. CFI = comparative fit index; TLI = Tucker–Lewis index; RMSEA = root-mean-square error of approximation; SRMR = standardized root-mean-square residual.

Table 7
Mean Levels of Depression, Anxiety, Stress, and GD

Scale	Range	Working-age cohort			Older cohort			<i>p</i>
		Total (<i>N</i> = 3,206)	Men (<i>N</i> = 640)	Women (<i>N</i> = 2,566)	Total (<i>N</i> = 5,736)	Men (<i>N</i> = 1,046)	Women (<i>N</i> = 4,690)	
Depression	0–21	3.5	3.9	3.4	3.3	3.2	3.3	.918
Anxiety	0–21	2.4	2.2	2.4	2.4	2.3	2.4	.040*
Stress	0–21	5.2	4.6	5.4	3.6	3.4	3.6	.503
Total (GD)	0–63	11.1	10.6	11.2	9.2	8.9	9.3	.196

Note. Intergender contrast *p* values derive from Welch's unequal variances *t* tests with a Bonferroni correction (*k* = 8). The data collected in 2022. GD = General Distress.
* *p* < .05.

Finally, the city of Helsinki itself lies within the largest urban area in Finland, the Helsinki metropolitan area, which hosts 28% of the country's population, the main transport hubs, and multiple universities. Naturally, the area differs from the rest of the country culturally, politically, and sociodemographically. Future users should consider whether our sample reflects their target population.

Most participants in both cohorts identified as women, which reflects the gender distribution of the target population (Lahelma et al., 2013; Lallukka et al., 2020). Analyses of statistical power showed that invariance analyses could be conducted by binary gender in both cohorts. Statistics Finland, the national statistical institution in Finland, defines foreign background as having a mother and father born outside of Finland. In 2022, 4% of the participants in the younger cohort reported having a mother and father born outside of Finland compared with 9% of the population of Finland and 19% of the population of the city of Helsinki (Statistics Finland, 2023). This discrepancy is in part explained by the decision to exclude answers in Swedish, English, and Russian. Still, more research is needed to validate the DASS-21 among Finnish speakers of other genders and Finns of rural or foreign background. In a cursory comparison, no significant difference was observed between the sample that answered in Finnish and the one that answered in other languages (Swedish, English, or Russian). Our analyses of interage group and binary interender invariance found no meaningful difference in either cohort, which supports the use of DASS-21 among Finnish-speaking Finns.

References

Aalto, A.-M., Aro, A. R., & Teperi, J. (1999). *RAND-36 terveyteen liittyvän elämänlaadun mittarina: Mittarin luotettavuus ja suomalaiset väestöarvot* [The RAND-36 as a measure of health-related quality of life: Measure reliability and Finnish population norms]. STAKES (Sosiaali- ja terveysalan tutkimus- ja kehittämiskeskus). <https://urn.fi/URN:NBN:fi-fe201211089642>

Ahmed, O., Faisal, R. A., Alim, S. M. A. H. M., Sharker, T., & Hiramoni, F. A. (2022). The psychometric properties of the Depression Anxiety Stress Scale-21 (DASS-21) Bangla version. *Acta Psychologica*, 223, Article 103509. <https://doi.org/10.1016/j.actpsy.2022.103509>

American Educational Research Association [AERA], American Psychological Association, & National Council on Measurement in Education. (2014). *Standards for educational and psychological testing*.

Antony, M. M., Bieling, P. J., Cox, B. J., Enns, M. W., & Swinson, R. P. (1998). Psychometric properties of the 42-item and 21-item versions of the Depression Anxiety Stress Scales in clinical groups and a community sample. *Psychological Assessment*, 10(2), 176–181. <https://doi.org/10.1037/1040-3590.10.2.176>

Beck, A. T., Epstein, N., Brown, G., & Steer, R. A. (1988). An inventory for measuring clinical anxiety: Psychometric properties. *Journal of Consulting and Clinical Psychology*, 56(6), 893–897. <https://doi.org/10.1037/0022-006X.56.6.893>

Bibi, A., Lin, M., Zhang, X. C., & Margraf, J. (2020). Psychometric properties and measurement invariance of Depression, Anxiety and Stress Scales (DASS-21) across cultures. *International Journal of Psychology*, 55(6), 916–925. <https://doi.org/10.1002/ijop.12671>

Caspi, A., Houts, R. M., Belsky, D. W., Goldman-Mellor, S. J., Harrington, H., Israel, S., Meier, M. H., Ramrakha, S., Shalev, I., Poulton, R., & Moffitt, T. E. (2014). The p factor: One general psychopathology factor in the structure of psychiatric disorders? *Clinical Psychological Science*, 2(2), 119–137. <https://doi.org/10.1177/2167702613497473>

Chin, E. G., Buchanan, E. M., Ebesutani, C., & Young, J. (2019). Depression, anxiety, and stress: How should clinicians interpret the total

- and subscale scores of the 21-item Depression, Anxiety, and Stress Scales? *Psychological Reports*, 122(4), 1550–1575. <https://doi.org/10.1177/003294118783508>
- Crawford, J. R., Cayley, C., Lovibond, P. F., Wilson, P. H., & Hartley, C. (2011). Percentile norms and accompanying interval estimates from an Australian general adult population sample for Self-Report Mood Scales (BAI, BDI, CRSD, CES-D, DASS, DASS-21, STAI-X, STAI-Y, SRDS, and SRAS). *Australian Psychologist*, 46(1), 3–14. <https://doi.org/10.1111/j.1742-9544.2010.00003.x>
- Crawford, J. R., & Henry, J. D. (2003). The Depression Anxiety Stress Scales (DASS): Normative data and latent structure in a large non-clinical sample. *British Journal of Clinical Psychology*, 42(2), 111–131. <https://doi.org/10.1348/014466503321903544>
- Daza, P., Novy, D. M., Stanley, M. A., & Averill, P. (2002). The Depression Anxiety Stress Scale-21: Spanish translation and validation with a Hispanic sample. *Journal of Psychopathology and Behavioral Assessment*, 24(3), 195–205. <https://doi.org/10.1023/A:1016014818163>
- de Gruijter, D. N. M., & van der Kamp, L. J. T. (2019). *Statistical test theory for the behavioral sciences*. CRC Press.
- Duffy, C. J., Cunningham, E. G., & Moore, S. M. (2005). Brief report: The factor structure of mood states in an early adolescent sample. *Journal of Adolescence*, 28(5), 677–680. <https://doi.org/10.1016/j.adolescence.2005.08.013>
- Finnish Institute for Health and Welfare. (2023, June 19). *Mielenterveyshäiriöt*. <https://thl.fi/fi/web/mielenterveys/mielenterveyshairiot>
- Global Burden of Disease 2019 Mental Disorders Collaborators. (2022). Global, regional, and national burden of 12 mental disorders in 204 countries and territories, 1990–2019: A systematic analysis for the Global Burden of Disease Study 2019. *The Lancet Psychiatry*, 9(2), 137–150. [https://doi.org/10.1016/S2215-0366\(21\)00395-3](https://doi.org/10.1016/S2215-0366(21)00395-3)
- Goodkind, M., Eickhoff, S. B., Oathes, D. J., Jiang, Y., Chang, A., Jones-Hagata, L. B., Ortega, B. N., Zaiko, Y. V., Roach, E. L., Korgaonkar, M. S., Grieve, S. M., Galatzer-Levy, I., Fox, P. T., & Etkin, A. (2015). Identification of a common neurobiological substrate for mental illness. *JAMA Psychiatry*, 72(4), 305–315. <https://doi.org/10.1001/jamapsychiatry.2014.2206>
- Hays, R. D., & Morales, L. S. (2001). The RAND-36 measure of health-related quality of life. *Annals of Medicine*, 33(5), 350–357. <https://doi.org/10.3109/07853890109002089>
- Henry, J. D., & Crawford, J. R. (2005). The short-form version of the Depression Anxiety Stress Scales (DASS-21): Construct validity and normative data in a large non-clinical sample. *British Journal of Clinical Psychology*, 44(2), 227–239. <https://doi.org/10.1348/014466505X29657>
- Hernández, A., Hidalgo, M. D., Hambleton, R. K., & Gómez-Benito, J. (2020). International Test Commission guidelines for test adaptation: A criterion checklist. *Psicothema*, 32(3), 390–398. <https://doi.org/10.7334/psicothema2019.306>
- Hu, L., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling*, 6(1), 1–55. <https://doi.org/10.1080/10705519909540118>
- International Test Commission. (2017). *The ITC guidelines for translating and adapting tests* (2nd ed.). <https://www.intestcom.org/>
- Joanes, D. N., & Gill, C. A. (1998). Comparing measures of sample skewness and kurtosis. *Statistician*, 47(1), 183–189. <https://doi.org/10.1111/1467-9884.00122>
- Jorgensen, T. D., Pornprasertmanit, S., Schoemann, A. M., & Rosseel, Y. (2022). *semTools: Useful tools for structural equation modeling*. <https://CRAN.R-project.org/package=semTools>
- Kotov, R., Krueger, R. F., Watson, D., Achenbach, T. M., Althoff, R. R., Bagby, R. M., Brown, T. A., Carpenter, W. T., Caspi, A., Clark, L. A., Eaton, N. R., Forbes, M. K., Forbush, K. T., Goldberg, D., Hasin, D., Hyman, S. E., Ivanova, M. Y., Lynam, D. R., Markon, K., ... Zimmerman, M. (2017). The hierarchical taxonomy of psychopathology (HiTOP): A dimensional alternative to traditional nosologies. *Journal of Abnormal Psychology*, 126(4), 454–477. <https://doi.org/10.1037/abn0000258>
- Lahelma, E., Aittomäki, A., Laaksonen, M., Lallukka, T., Martikainen, P., Piha, K., Rahkonen, O., & Saastamoinen, P. (2013). Cohort profile: The Helsinki Health Study. *International Journal of Epidemiology*, 42(3), 722–730. <https://doi.org/10.1093/ije/dys039>
- Lahti, J., Knop, J., Lallukka, T., Harkko, J., & Kouvonen, A. (2023). Occupational class differences in emotional exhaustion among municipal employees—The role of employment sector and psychosocial working conditions. *Psychological Reports*, 126(6), 3104–3122. <https://doi.org/10.1177/00332941221106393>
- Lallukka, T., Pietiläinen, O., Jäppinen, S., Laaksonen, M., Lahti, J., & Rahkonen, O. (2020). Factors associated with health survey response among young employees: A register-based study using online, mailed and telephone interview data collection methods. *BMC Public Health*, 20(1), Article 184. <https://doi.org/10.1186/s12889-020-8241-8>
- Lee, J., Lee, E.-H., & Moon, S. H. (2019). Systematic review of the measurement properties of the Depression Anxiety Stress Scales-21 by applying updated COSMIN methodology. *Quality of Life Research*, 28(9), 2325–2339. <https://doi.org/10.1007/s11136-019-02177-x>
- Li, C.-H. (2016). Confirmatory factor analysis with ordinal data: Comparing robust maximum likelihood and diagonally weighted least squares. *Behavior Research Methods*, 48(3), 936–949. <https://doi.org/10.3758/s13428-015-0619-7>
- Lovibond, P. F. (2022). *Depression Anxiety Stress Scales—DASS: Frequently asked questions*. <https://www2.psy.unsw.edu.au/groups/dass>
- Lovibond, P. F., & Lovibond, S. H. (1995). The structure of negative emotional states: Comparison of the Depression Anxiety Stress Scales (DASS) with the Beck Depression and Anxiety Inventories. *Behaviour Research and Therapy*, 33(3), 335–343. [https://doi.org/10.1016/0005-7967\(94\)00075-U](https://doi.org/10.1016/0005-7967(94)00075-U)
- Newman, D. L., Moffitt, T. E., Caspi, A., & Silva, P. A. (1998). Comorbid mental disorders: Implications for treatment and sample selection. *Journal of Abnormal Psychology*, 107(2), 305–311. <https://doi.org/10.1037/0021-843X.107.2.305>
- Olsson, U. (1979). Maximum likelihood estimation of the polychoric correlation coefficient. *Psychometrika*, 44(4), 443–460. <https://doi.org/10.1007/BF02296207>
- Osman, A., Wong, J. L., Bagge, C. L., Freedenthal, S., Gutierrez, P. M., & Lozano, G. (2012). The Depression Anxiety Stress Scales-21 (DASS-21): Further examination of dimensions, scale reliability, and correlates. *Journal of Clinical Psychology*, 68(12), 1322–1338. <https://doi.org/10.1002/jclp.21908>
- R Core Team. (2022). *R: A language and environment for statistical computing* [Computer software]. R Foundation for Statistical Computing. <https://www.R-project.org/>
- Raiche, G., & Magis, D. (2022). *nFactors: Parallel analysis and other non graphical solutions to the Cattell scree test*. <https://CRAN.R-project.org/package=nFactors>
- Räsänen, P., Lappalainen, P., Muotka, J., Tolvanen, A., & Lappalainen, R. (2016). An online guided ACT intervention for enhancing the psychological wellbeing of university students: A randomized controlled clinical trial. *Behaviour Research and Therapy*, 78, 30–42. <https://doi.org/10.1016/j.brat.2016.01.001>
- Revelle, W. (2023). *psych: Procedures for psychological, psychometric, and personality research_* (Version R package Version 2.3.3) [Computer software]. Northwestern University. <https://CRAN.R-project.org/package=psych>
- Rosenström, T., Gjerde, L. C., Krueger, R. F., Aggen, S. H., Czajkowski, N. O., Gillespie, N. A., Kendler, K. S., Reichborn-Kjennerud, T., Torvik, F. A., & Ystrom, E. (2019). Joint factorial structure of psychopathology and personality. *Psychological Medicine*, 49(13), 2158–2167. <https://doi.org/10.1017/S0033291718002982>

- Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, 48(2), 1–36. <https://doi.org/10.18637/jss.v048.i02>
- Scholten, S., Velten, J., Bieda, A., Zhang, X. C., & Margraf, J. (2017). Testing measurement invariance of the Depression, Anxiety, and Stress Scales (DASS-21) across four countries. *Psychological Assessment*, 29(11), 1376–1390. <https://doi.org/10.1037/pas0000440>
- Shankman, S. A., & Klein, D. N. (2003). The relation between depression and anxiety: An evaluation of the tripartite, approach-withdrawal and valence-arousal models. *Clinical Psychology Review*, 23(4), 605–637. [https://doi.org/10.1016/S0272-7358\(03\)00038-2](https://doi.org/10.1016/S0272-7358(03)00038-2)
- Sireci, S., & Benítez, I. (2023). Evidence for test validation: A guide for practitioners. *Psicothema*, 35(3), 217–226. <https://doi.org/10.7334/psicothema2022.477>
- Sourander, A., & Marttunen, M. (2016). Lasten ja nuorten mielenterveyden häiriöiden epidemiologia [Epidemiology of mental health disorders among children and adolescents]. In K. Kumpulainen (Ed.), *Lastenpsykiatria ja nuorisopsykiatria* (pp. 116–123). Kustannus Oy Duodecim.
- Statistics Finland. (2023). *Official statistics of Finland (OSF): Population structure*. <https://stat.fi/en/statistics/vaerak>
- Tavakol, M., & Dennick, R. (2011). Making sense of Cronbach's alpha. *International Journal of Medical Education*, 2, 53–55. <https://doi.org/10.5116/ijme.4dfb.8dfd>
- Thériault, R. (2023). rempsyc: Convenience functions for psychology. *Journal of Open Source Software*, 8(87), Article 5466. <https://doi.org/10.21105/joss.05466>
- Tran, T. D., Tran, T., & Fisher, J. (2013). Validation of the Depression Anxiety Stress Scales (DASS) 21 as a screening instrument for depression and anxiety in a rural community-based cohort of northern Vietnamese women. *BMC Psychiatry*, 13(1), Article 24. <https://doi.org/10.1186/1471-244X-13-24>
- Tully, P. J., Zajac, I. T., & Venning, A. J. (2009). The structure of anxiety and depression in a normative sample of younger and older Australian adolescents. *Journal of Abnormal Child Psychology*, 37(5), 717–726. <https://doi.org/10.1007/s10802-009-9306-4>
- United Nations Educational, Scientific and Cultural Organization Institute for Statistics. (2012). *International Standard Classification of Education*. <https://uis.unesco.org/sites/default/files/documents/international-standard-classification-of-education-isced-2011-en.pdf>
- Vähäsarja, L. E. J. (2024). *Code for "Depression Anxiety Stress Scales (DASS-21) for Finnish-speaking adults: Validation and normative data"* [Dataset]. <https://doi.org/10.17605/OSF.IO/6Q8FK>
- van de Schoot, R., Lugtig, P., & Hox, J. (2012). A checklist for testing measurement invariance. *European Journal of Developmental Psychology*, 9(4), 486–492. <https://doi.org/10.1080/17405629.2012.686740>
- Wang, K., Gaitsch, H., Poon, H., Cox, N. J., & Rzhetsky, A. (2017). Classification of common human diseases derived from shared genetic and environmental determinants. *Nature Genetics*, 49(9), 1319–1325. <https://doi.org/10.1038/ng.3931>
- Wilcosky, T., & Wing, S. (1987). The healthy worker effect. Selection of workers and work forces. *Scandinavian Journal of Work, Environment & Health*, 13(1), 70–72. <https://doi.org/10.5271/sjweh.2078>
- Winer, E. S., Bryant, J., Bartoszek, G., Rojas, E., Nadorff, M. R., & Kilgore, J. (2017). Mapping the relationship between anxiety, anhedonia, and depression. *Journal of Affective Disorders*, 221, 289–296. <https://doi.org/10.1016/j.jad.2017.06.006>
- Zanon, C., Brenner, R. E., Baptista, M. N., Vogel, D. L., Rubin, M., Al-Darmaki, F. R., Gonçalves, M., Heath, P. J., Liao, H.-Y., Mackenzie, C. S., Topkaya, N., Wade, N. G., & Zlati, A. (2021). Examining the dimensionality, reliability, and invariance of the Depression, Anxiety, and Stress Scale-21 (DASS-21) across eight countries. *Assessment*, 28(6), 1531–1544. <https://doi.org/10.1177/1073191119887449>
- Zhang, Z., & Mai, Y. (2023). *WebPower: Basic and advanced statistical power analysis*. <https://CRAN.R-project.org/package=WebPower>
- Zinbarg, R. E., Revelle, W., Yovel, I., & Li, W. (2005). Cronbach's α , Revelle's β , and McDonald's ω H: Their relations with each other and two alternative conceptualizations of reliability. *Psychometrika*, 70(1), 123–133. <https://doi.org/10.1007/s11336-003-0974>

Received February 29, 2024

Revision received September 30, 2024

Accepted October 15, 2024 ■